

Integration of Power BI with SharePoint Online, PowerApps and Flow

ABOUT



In professional life consultant, Microsoft Certified Trainer, administrator, developer, freelancer, sometimes a miracle-worker and magician.



From the beginning of career associated with Microsoft technologies.



Private life in numbers: 1 husband, 1 daughter, 1 cat and 2 dogs. My hobbies are detective stories and photography.



Email: info@zalnet.pl



Skype: beata.zalewa



WWW: <https://zalnet.pl/> <https://zalnet.pl/blog>

AGENDA



Intro



SPO



PowerApps



Flow



Power BI
Desktop



Power BI



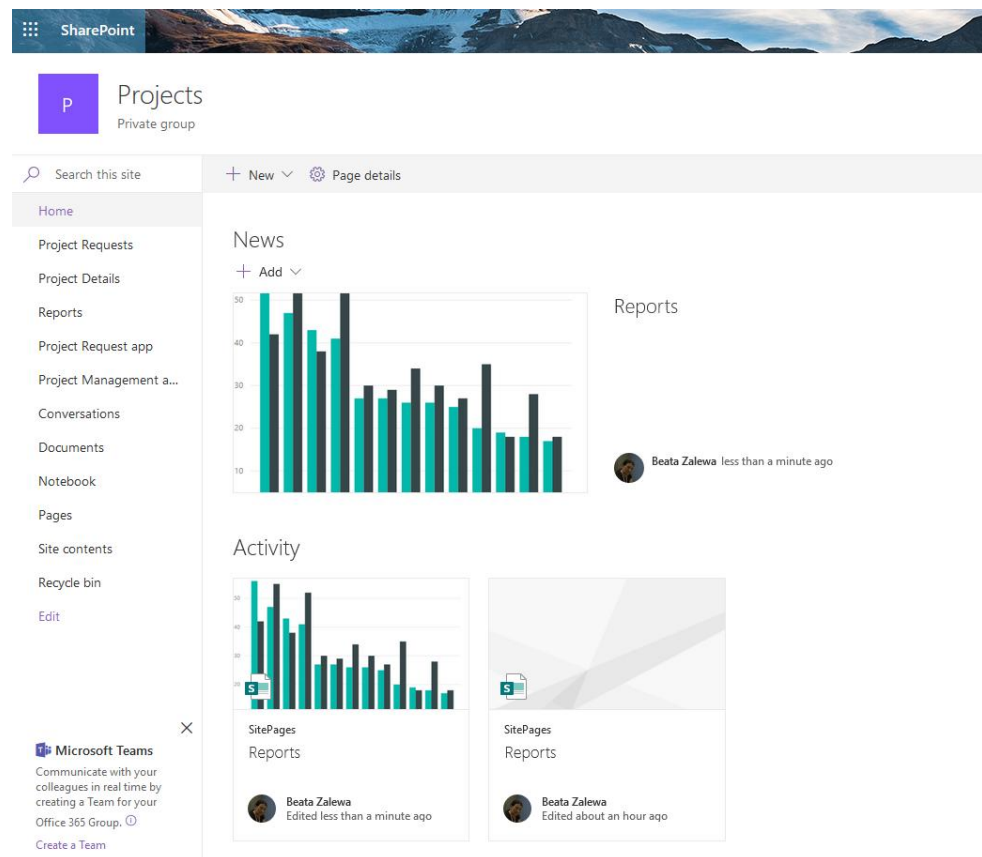
Publish
reports



Host
reports

INTRO

- Do you have SharePoint Online and want to better automate and streamline your business processes?
- Have you worked with **PowerApps**, **Microsoft Flow**, or **Power BI**, but you're not sure how to use them with **SharePoint Online**?
- You will see how to use basic PowerApps app for project management based on SharePoint lists and three key technologies that integrate with SharePoint Online: PowerApps, Microsoft Flow, and Power BI.
- These technologies work together, making it easy to measure your business, act on the results, and automate your workflows.



SharePoint Online

SharePoint Online is a cloud-based service that helps organizations share and manage content, knowledge, and applications to:

- Empower teamwork
- Quickly find information
- Seamlessly collaborate across the organization

SharePoint

P

Projects

Private group

Search

Home

Conversations

Documents

Shared with us

Notebook

Pages

Project Requests

Project Details

Site contents

Project Management a...

Project Request app

Recycle bin

Edit

New

Quick edit

Export to Excel

PowerApps

Flow

Project Requests

Title	Description	ProjectType	RequestDate	Requestor	EstimatedDays	Approved
Mobile devices for sales team	The sales team will now use Contoso-supplied devices	New hardware	September 19, 2016	Zoe Watson	30	Yes
Monitor upgrade - finance department	All members of finance will move from 19-inch to 24-inch monitors	Hardware update	January 23, 2016	Zoe Watson	15	Yes
New version of CRM system	Upgrade to version 7.4, based on new features	Software update	December 9, 2016	Steve Smith	30	Yes
New tablets for designers	The design team is moving to a new brand of tablet	Hardware update	September 8, 2016	Penny Coventry	20	Yes
Mobile devices for HR team	The HR team will now use Contoso-supplied devices	New hardware	November 2, 2016	Zoe Watson	45	Yes

List in SharePoint Online

A list in SharePoint is a collection of data that gives you and your co-workers a flexible way to organize information.

We can create new content in list in modern or classic manner.

The image shows a SharePoint Online interface. On the left, a table with two columns: 'EstimatedDays' and 'Approved'. The 'EstimatedDays' column has values 30, 15, 30, 20, and 45. The 'Approved' column has values Yes, Yes, Yes, Yes, and Yes. To the right of the table is a '+ Add column' button. On the right, the 'Edit column' dialog is open. It has a title bar with a close button (X). The dialog contains the following fields and options:

- Edit column** (Title)
- [Learn more about column types and options.](#) (Link)
- Name *** (Text input): Approved
- Description** (Text input): (Empty)
- Type** (Dropdown menu): Single line of text
- Default value** (Text input): Enter a default value
- ☐ Use calculated value ⓘ
- [More options](#) (Link)
- Buttons:** Save (purple), Cancel (grey), Delete (grey)

SharePoint Online

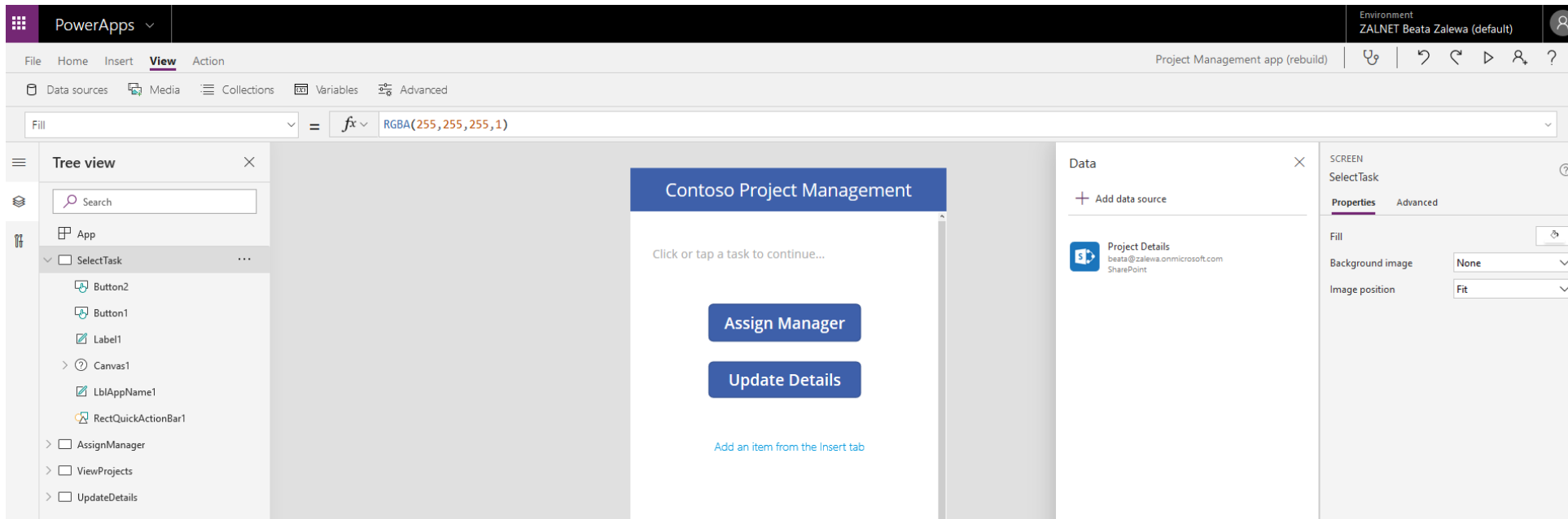
Once your list is created, add columns for your content.

Add new columns by selecting + Add Column on the far right side of the column headers.

DEMO 1

SharePoint Online

<https://zalewa.sharepoint.com/sites/Projects>



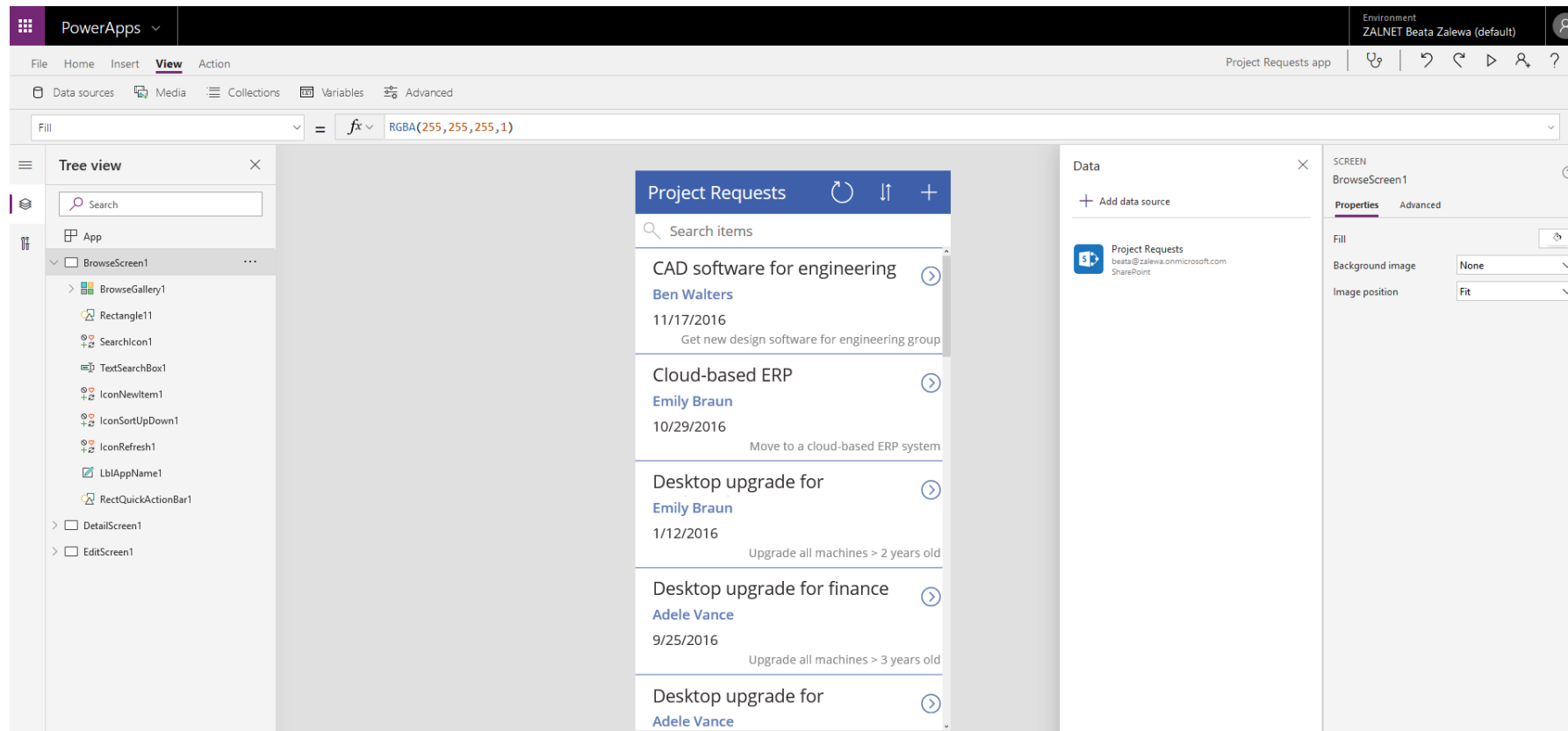
PowerApps in SharePoint Online

With Power Apps, Microsoft has provided a platform to create custom applications (Mobile and Desktop).

Requiring no heavy coding or experience, you can develop them to meet a variety of internal needs.

Similarly, you can link your existing data sources into Power Apps and create one-of-a-kind applications.

With the option to link to over 200 sources and 100 templates, this provides you with what you want.



PowerApps

PowerApps has connections to various Microsoft-based and third-party applications, including Microsoft SQL Server, SharePoint and OneDrive in the Office 365 productivity suite, as well as Dropbox, Google Docs, SAP, Oracle and Dynamics CRM.

 Assign Manager

Select a project, then assign a manager

Mobile devices for engineering
Not started
9/27/2016

New BI software
Not started
5/12/2016

Manager:

Using PowerApps

DEMO 2

PowerApps

<https://create.powerapps.com/studio/#>



Michał Guzowski ► Microsoft 365 User Group Poland

25 min · 🌐



Znowu coś siadło.

**Apki PowerApps mi się nie
odpalają.**

**A na niektórych tenantach
SPO mam read-only.**



Projects

Private group

Search

Home

Conversations

Documents

Shared with us

Notebook

Pages

Project Requests

Project Details

Site contents

+ New

Quick edit

Export to Excel

PowerApps

Flow

...

Project Requests

Title	Description		RequestDate	Requestor	EstimatedDays	Approved
Mobile devices for sales team	The sales team will now use Contoso-supplied devices	New hardware	September 19, 2016	Emily Braun	30	Yes
Monitor upgrade - finance department	All members of finance will move from 19-inch to 24-inch monitors	Hardware update	January 23, 2016	Emily Braun	15	Yes
New version of CRM system	Upgrade to version 7.4, based on new features	Software update	December 9, 2016	Ben Walters	30	Yes

Set a reminder



Create a flow

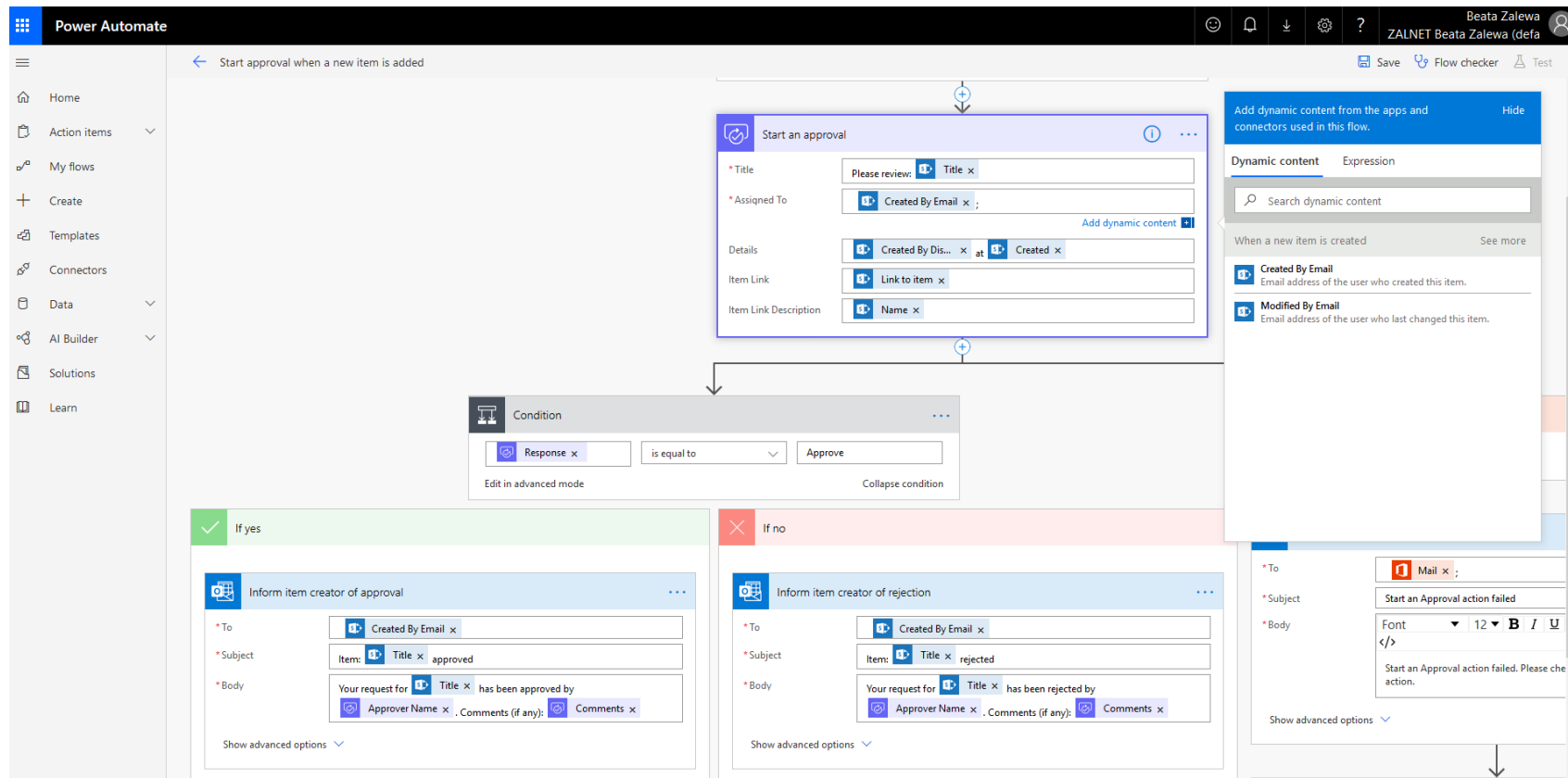
See your flows

Configure flows

Flow

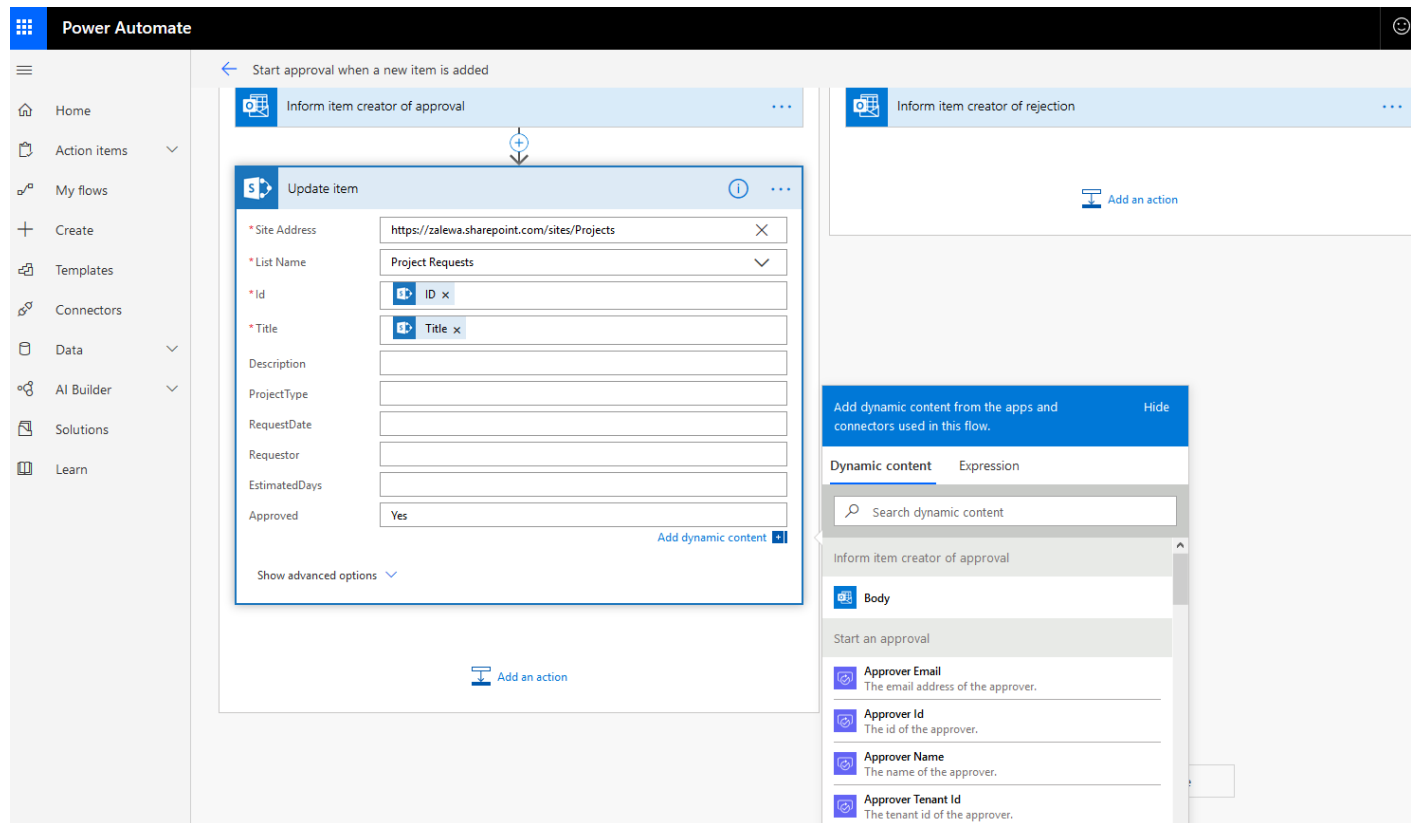
Microsoft Flow (now Power Automate) is cloud-based software that allows employees to create and automate workflows and tasks across multiple applications and services without help from developers.

Automated workflows are called *flows*. To create a flow, the user specifies what action should take place when a specific event occurs.



Flow

Once a flow is built, it can be managed on the desktop or through an app on a mobile device. Flow integrates with various Microsoft services and apps, including Power BI, PowerApps, Dynamics 365 and Office 365.



Flow

Actions open a range of possibilities for composing business logic.

With Actions, you can perform operations, such as Create, Update, Delete, Assign, or Perform Action.

Internally, an action creates a custom message.

The screenshot displays the Microsoft Flow 'Update item' action configuration. The main configuration pane on the left includes the following fields:

- * Site Address: <https://zalewa.sharepoint.com/sites/Projects>
- * List Name: Project Requests
- * Id: ID (with a dynamic content icon and 'Add dynamic content' link)
- * Title: (empty text box)
- Description: (empty text box)
- ProjectType: (empty text box)
- RequestDate: (empty text box)
- Requestor: (empty text box)
- EstimatedDays: (empty text box)
- Approved: (empty text box)

The right pane shows the 'Dynamic content' section with a search bar and a list of dynamic content items:

- Inform item creator of approval
- Body
- When a new item is created
- ID: List item id. Use this value for specifying the item to act on in...

Flow

Developers refer to these actions as **messages**.

Each of these messages is based on actions taken on an entity record. If the goal of a process is to create a record, then update it, and then assign it, there are three separate steps.

Each step is defined by the capabilities of the entity – not necessarily your business process.

SaveCancelCopy linkCustomize with PowerApps

New Monitor for Megan

Title *

New Monitor for Megan

Description

Megan needs a 24" monitor

ProjectType

New hardware

RequestDate

11/7/2019

Requestor

Megan Markle

EstimatedDays

1

Approved

Pending

Attachments

[Add attachments](#)

SaveCancel

Flow result

To test the flow we must add new item to our list .

zalnet

✓

Focused

Other

Filter

Microsoft Flow

Please review: I need a new la... 05:24

BZ Requested by Beata Zalewa <beata@zalewa...>

Microsoft Flow

Please review: New Monitor fo... 05:22

BZ Requested by Beata Zalewa <beata@zalewa...>

Projects

You've joined the Projects group 04:05

Work Brilliantly Together Welcome to the Proje...

Last month

Windows Insider Program

Our gift to you Wed 16/10

The latest preview feature highlights, tips and t...

September

SharePoint Online

Uwaga! Zauważyliśmy, że nied... 22/09/2019

Please review: New Monitor for Megan

Getting too much email? [Unsubscribe](#)

Attachments, pictures, and links in this message have been blocked because the sender isn't in your Safe Senders list. [I trust content from maccount@microsoft.com.](#) | [Show blocked content](#)

Microsoft Flow <maccount@microsoft.com>

Thu 07/11/2019 05:22

Beata Zalewa

Approvals

Powered by Microsoft Flow

Pending approval

Requested by: Beata Zalewa [beata@zalewa.onmicrosoft.com](#)

Date Created: Thursday, November 7, 2019 4:22 AM GMT

Details: Beata Zalewa at Thursday, November 7, 2019 4:17 AM GMT

Link: [New Monitor for Megan](#)

Approve

Reject

Get the Flow app to receive push notifications and grant approvals from anywhere. [Learn more](#)

This message was created by an automated workflow in Microsoft Flow. Do not reply. Microsoft Corporation 2019.

Flow result

Flow sent an email as expected.

zalnet

Please review: New Monitor for Megan

Getting too much email? [Unsubscribe](#)

Attachments, pictures, and links in this message have been blocked because the sender isn't in your Safe Senders list. [I trust content from maccount@microsoft.com.](#) | [Show blocked content](#)

 Microsoft Flow <maccount@microsoft.com>
Thu 07/11/2019 05:22
Beata Zalewa

 Approvals | Powered by Microsoft Flow

Pending approval

Requested by: Beata Zalewa beata@zalewa.onmicrosoft.com
Date Created: Thursday, November 7, 2019 4:22 AM GMT
Details: Beata Zalewa at Thursday, November 7, 2019 4:17 AM GMT
Link: [New Monitor for Megan](#)

She is our CEO.

Get the Flow app to receive push notifications and grant approvals from anywhere. [Learn more](#)
This message was created by an automated workflow in Microsoft Flow. Do not reply. Microsoft Corporation 2019.

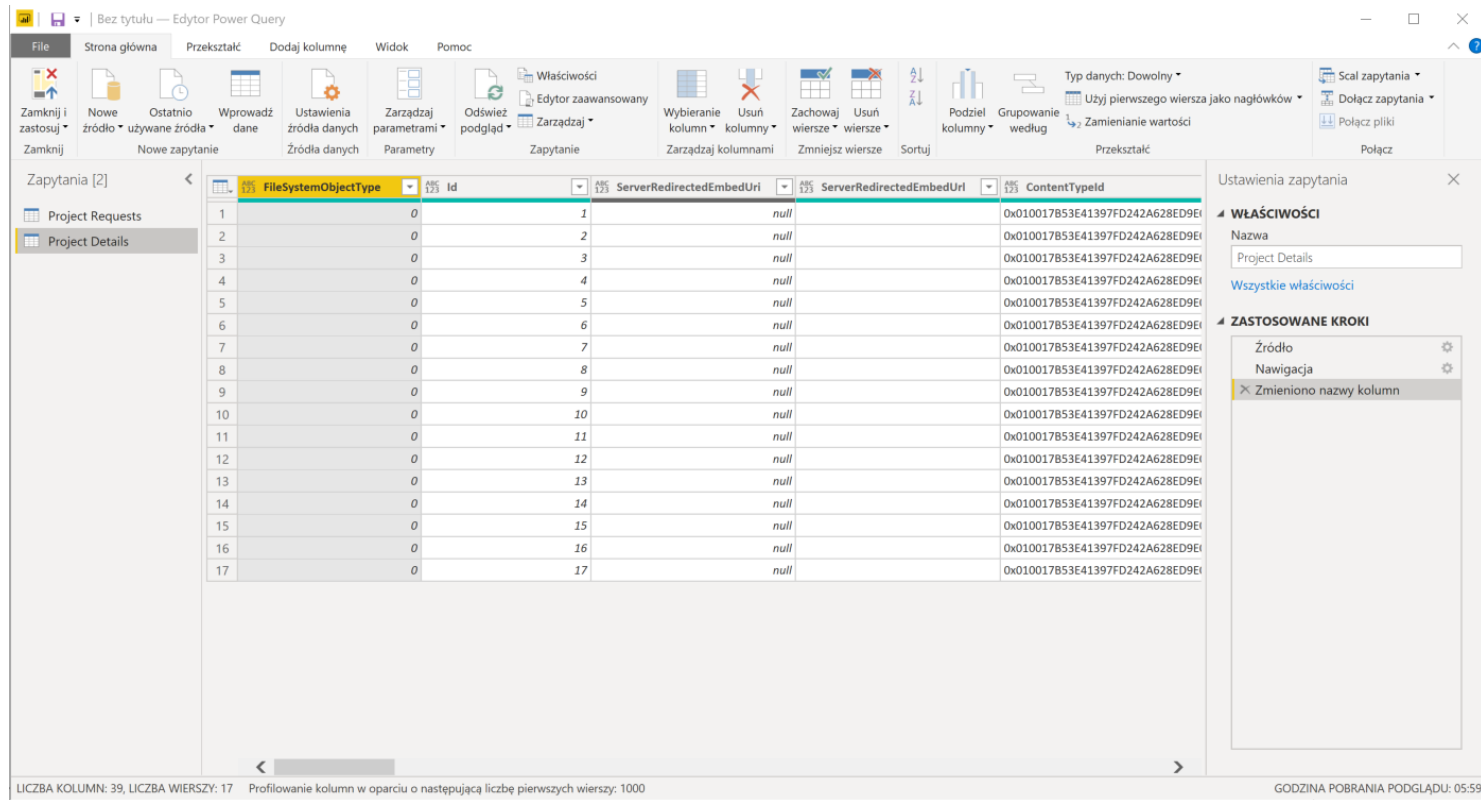
Flow result

We can approve or reject the request from our email level.

DEMO 3

Power Automate (Microsoft Flow)

<https://emea.flow.microsoft.com>



Power BI Desktop

Also known as **Power BI Free**, Power BI Desktop is a free desktop application you can install right on your own computer.

Power BI Desktop works cohesively with the Power BI service by providing advanced data exploration, shaping, modeling, and report creation with highly interactive visualizations. You can save your work to a file, and publish your data and reports right to your Power BI site to share with others.

Power BI Free is included in all Office 365 Plans, and you can sign up for Power BI Free any time you like.

Bez tytułu — Edytor Power Query

File Strona główna Przekształć Dodaj kolumnę Widok Pomoc

Zamknij i zastosuj Zamknij Nowe źródło Ostatnio używane źródła Wprowadź dane Ustawienia źródła danych Zarządzaj parametrami Odśwież podgląd Właściwości Edytor zaawansowany Zarządzaj Wybieranie kolumn Usun kolumny Zachowaj wiersze Usun wiersze Podziel kolumny Grupowanie według Typ danych: Dowolny Użyj pierwszego wiersza jako nagłówków Zamienianie wartości Scał zapytania Dołącz zapytania Połącz pliki

Zapytania [2]

- Project Requests
- Project Details

	ApprovedDate	Status	ProjectedStartDate	ProjectedEndDate	ProjectedDays	ActualDays
1	22-09-2016	Completed	10-11-2016	11-07-2016 07:00:00	20	
2	02-03-2016	Completed	02-11-2016	03-04-2016 07:00:00	17	
3	17-10-2016	Completed	25-10-2016	20-12-2016 08:00:00	41	
4	15-08-2016	Completed	09-12-2016	17-10-2016 07:00:00	26	
5	15-02-2016	Completed	03-01-2016	05-04-2016 07:00:00	47	
6	30-11-2016	Completed	12-05-2016	01-10-2017 07:00:00	27	
7	20-10-2016	Completed	28-10-2016	27-12-2016 08:00:00	43	
8	11-11-2016	In progress	01-09-2017	23-01-2017 08:00:00	11	
9	15-06-2016	Completed	20-06-2016	09-05-2016 07:00:00	56	
10	06-06-2016	Completed	15-06-2016	20-07-2016 07:00:00	26	
11	27-09-2016	Not started	null	null	null	
12	12-05-2016	Not started	null	null	null	
13	10-04-2016	Completed	10-10-2016	11-03-2016 08:00:00	19	
14	11-10-2016	Completed	18-11-2016	13-12-2016 08:00:00	18	
15	01-12-2016	Completed	18-01-2016	19-02-2016 08:00:00	25	
16	15-11-2016	Completed	22-11-2016	28-12-2016 08:00:00	27	
17	07-11-2019	In progress	07-11-2019	null	null	

Ustawienia zapytania

WŁAŚCIWOŚCI

Nazwa

Project Details

Wszystkie właściwości

ZASTOSOWANE KROKI

- Źródło
- Nawigacja
- Zmieniono nazwy kolumn
- Usunięto kolumny
- Zmieniono typ

Edit column in a report

You can use SharePoint list as a data source for Power BI report.

File Strona główna Modelowanie Pomoc

Zarządzaj relacjami Relacje Nowa miara Nowa kolumna Nowa tabela Nowy parametr Sortuj według kolumny Formatowanie Typ danych: Format: \$ % + .00 Auto Tabela główna: Kategoria danych: Brak kategorii Sumowanie domyślne: Nie sumuj Właściwości Zarządzaj rolami Wyświetl jako role Zabezpieczenia Nowa grupa Edytuj grupy Oznacz jako tabelę dat Kalendarze Język Schemat językowy Pytania i odpowiedzi

Id	Title	RequestID	ApprovedDate	Status	ProjectedStartDate	ProjectedEndDate	ProjectedDays	ActualDays	PMAssigned
1	Mobile devices for sales team	1	22 September 2016	Completed	10 November 2016	11 July 2016	20	35	Caroline Lee
2	Monitor upgrade - finance department	2	02 March 2016	Completed					
3	New version of CRM system	3	17 October 2016	Completed					
4	New tablets for designers	4	15 August 2016	Completed					
5	Mobile devices for HR team	5	15 February 2016	Completed					
6	CAD software for engineering	6	30 November 2016	Completed					
7	Mobile devices for finance team	7	20 October 2016	Completed					
8	Cloud-based ERP	8	11 November 2016	In progress					
9	New version of productivity software	9	15 June 2016	Completed					
10	Monitor upgrade - sales	10	06 June 2016	Completed					
11	Mobile devices for engineering team	11	27 September 2016	Not started					
12	New BI software	12	12 May 2016	Not started					
13	Desktop upgrade for finance	14	10 April 2016	Completed					
14	Desktop upgrade for marketing	15	11 October 2016	Completed					
15	Mobile devices for analytics team	17	01 December 2016	Completed					
16	New accounting package	18	15 November 2016	Completed					
17	New Monitor for Megan	22	07 November 2019	In progress					

Tworzenie relacji

Wybierz powiązane tabele i kolumny.

Project Requests

Id	Title	Description	ProjectType	Re
2	Mobile devices for sales team	The sales team will now use Contoso-supplied devices	New hardware	19 5
3	Monitor upgrade - finance department	All members of finance will move from 19-inch to 24-in...	Hardware update	2
4	New version of CRM system	Upgrade to version 7.4, based on new features	Software update	09

Project Details

Id	Title	RequestID	ApprovedDate	Status	ProjectedStartDate	Projecte
1	Mobile devices for sales team	1	22 September 2016	Completed	10 November 2016	
2	Monitor upgrade - finance department	2	02 March 2016	Completed	02 November 2016	
3	New version of CRM system	3	17 October 2016	Completed	25 October 2016	20 D

Kardynalność: Jeden do jednego (1:1)

Kierunek filtrowania krzyżowego: Oba

☒ Aktywuj tę relację ☐ Zastosuj filtr zabezpieczeń w obu kierunkach

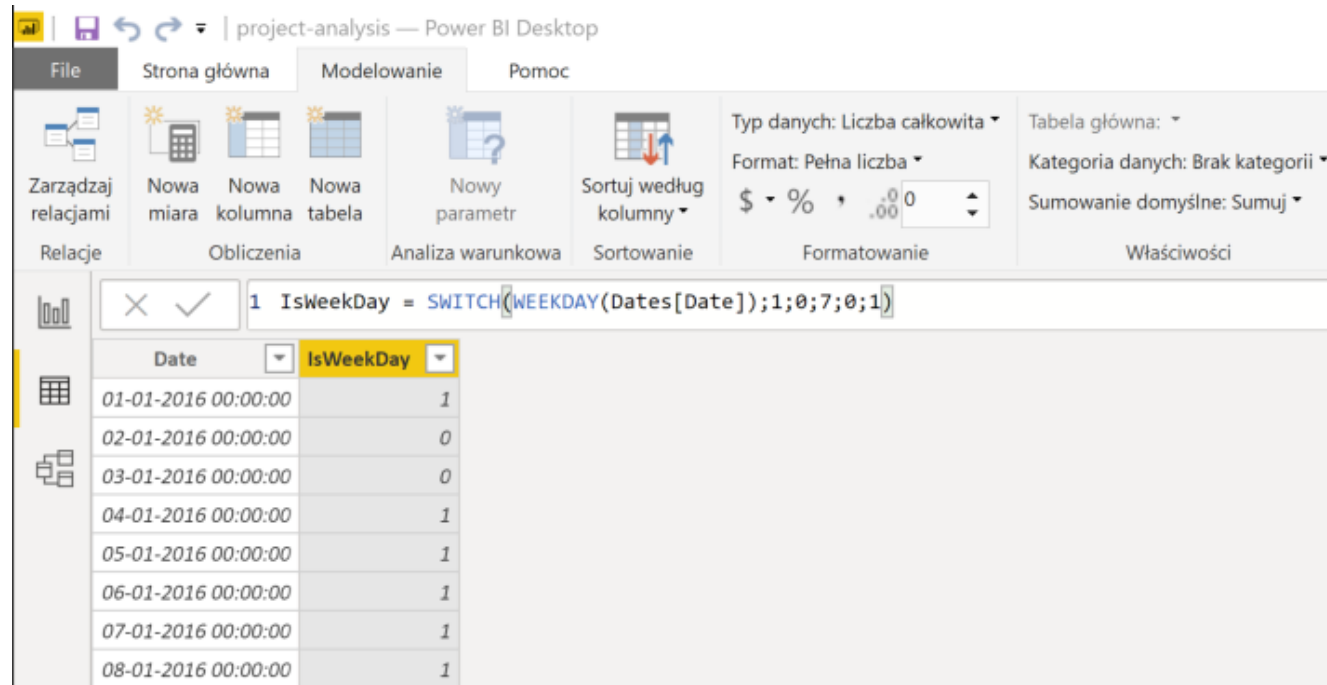
☐ Przyjmowanie integralności referencyjnej

OK Anuluj

Add relationships to Power BI tables

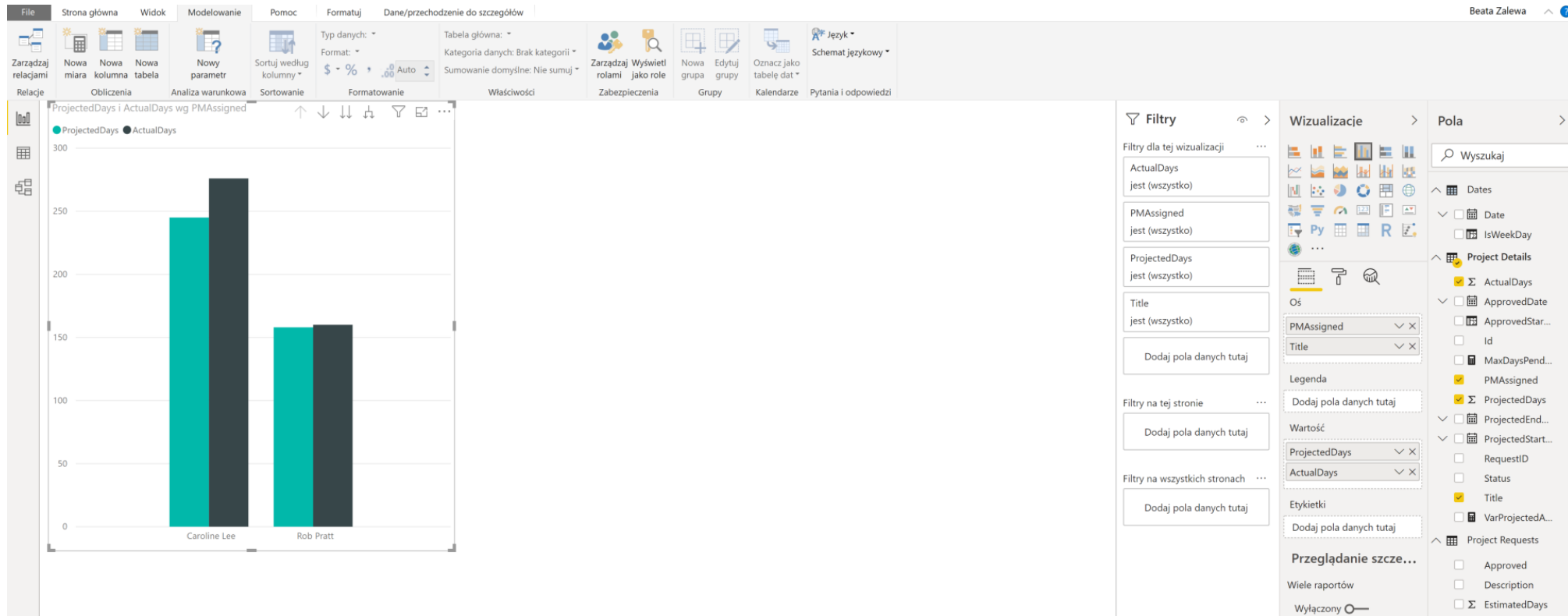
Relationship in Power BI is used to define the connections or the relation between two or more tables.

The relationship is used when we want to perform an analysis based on multiple tables.



Create new table and columns

We can create new tables using DAX expressions.

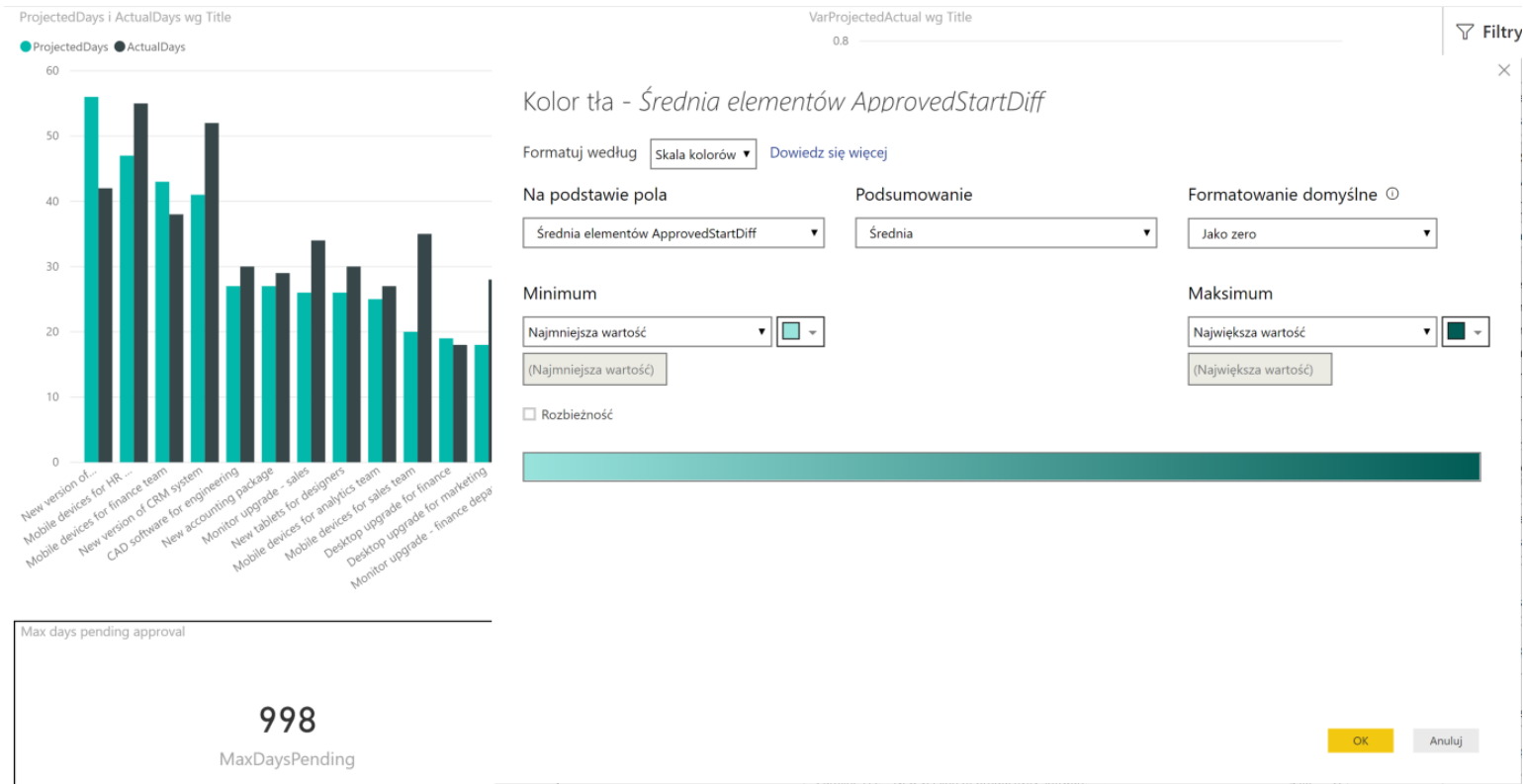


Create reports

Power BI Desktop includes Report View, where you can create any number of report pages with visualizations.

Report View in provides pretty much the same design experience as a report's Editing View in the Power BI service.

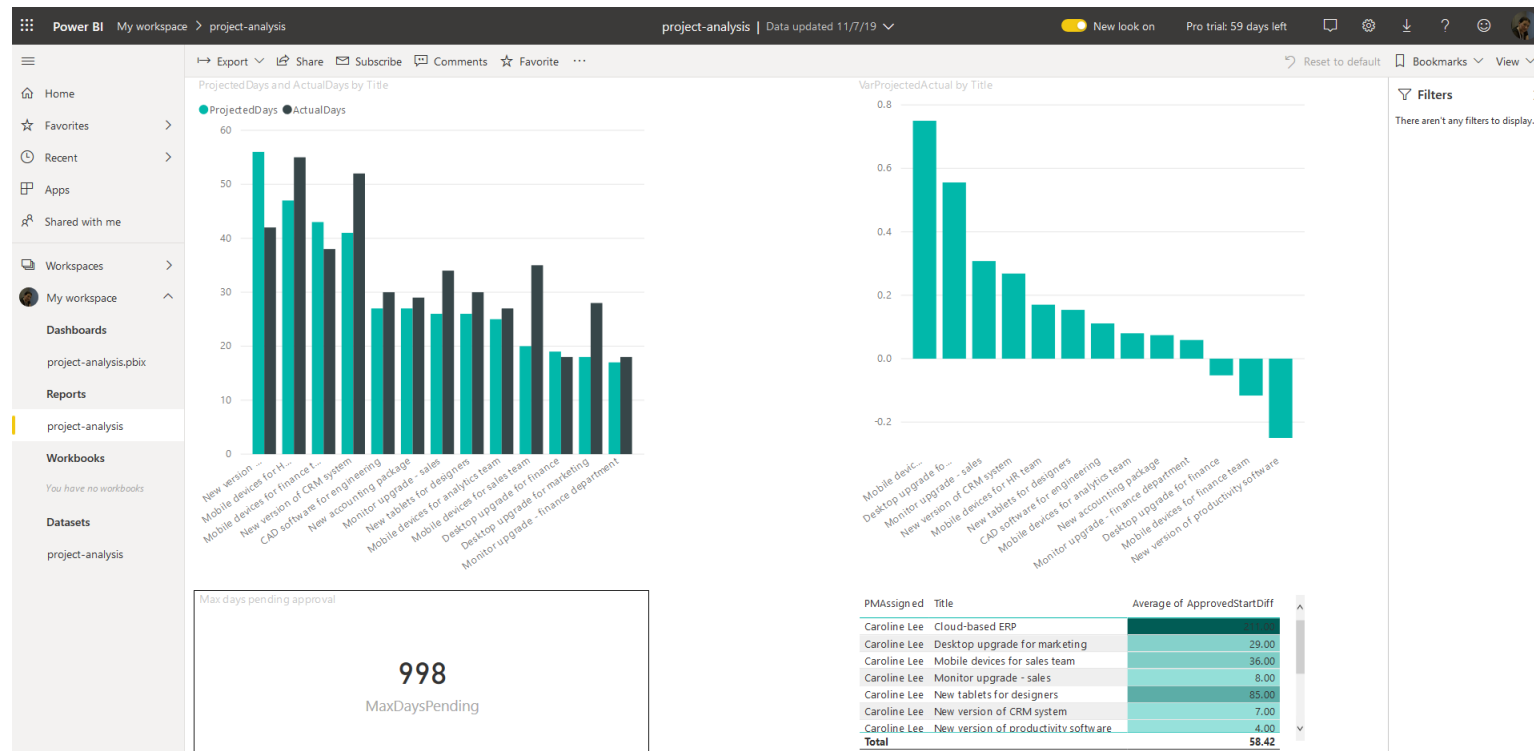
You can move visualizations around, copy and paste, merge, and so on.



Conditional formatting

Conditional formatting is a feature in Power BI and Excel that allows us to specify customized cell colours based on cell values, other values or fields by using gradient colours.

You can also display cell values with data bars in order to easily detect patterns and insights from the numbers.



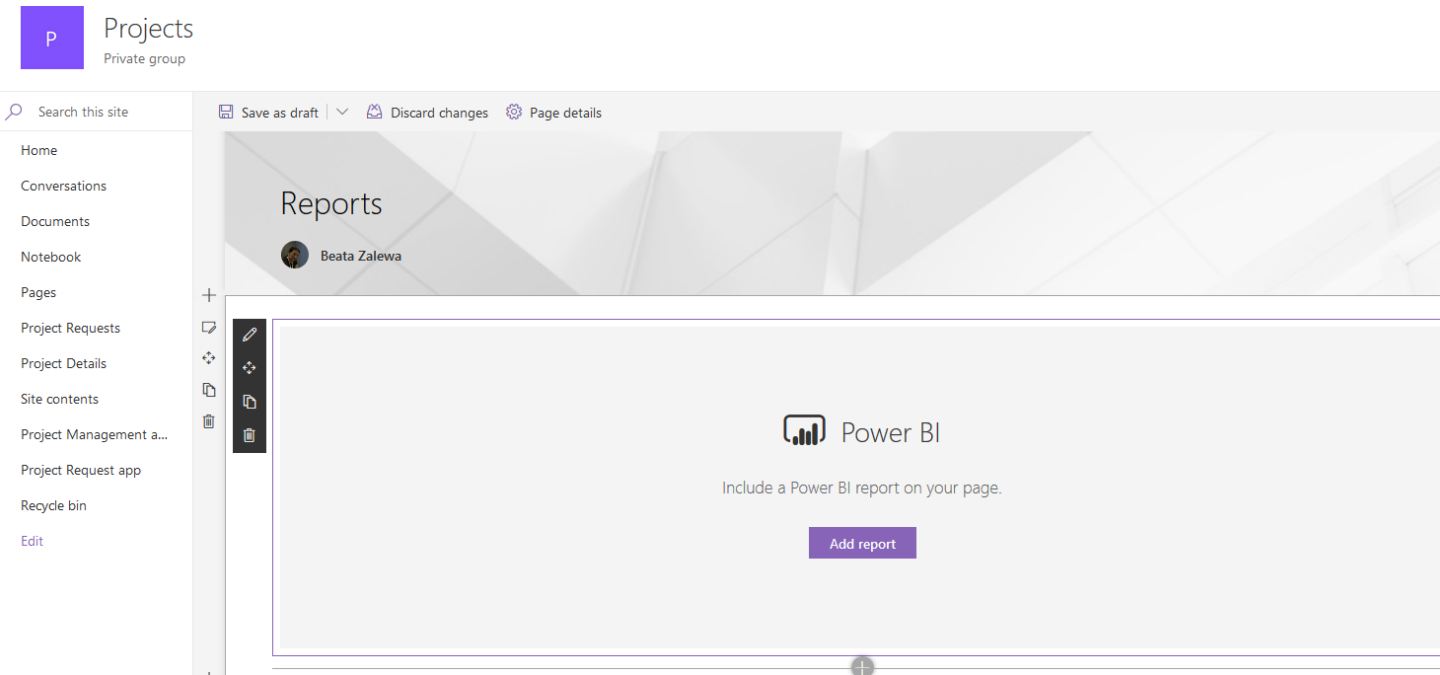
Formatting result

After creating all reports we can publish them on SharePoint page.

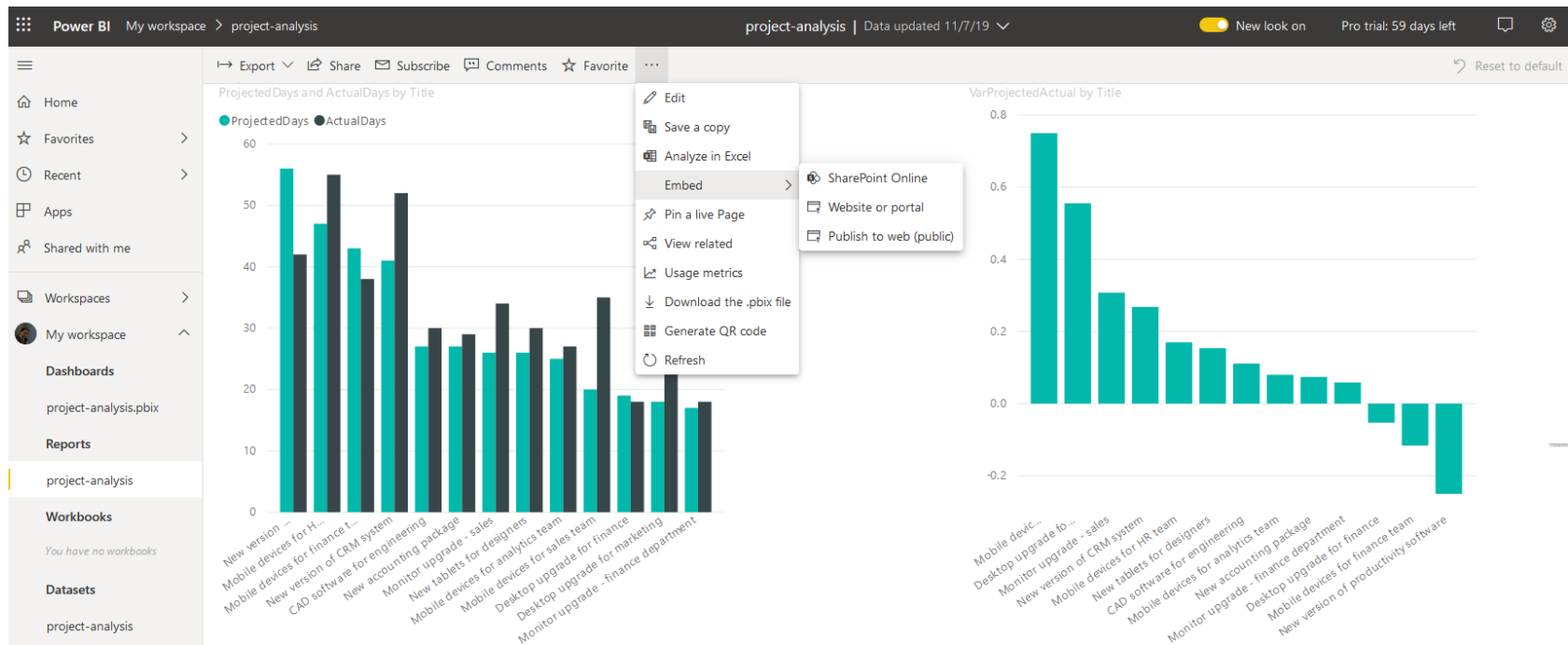
DEMO 4

Power BI

<https://app.powerbi.com>



Create SharePoint page for reports



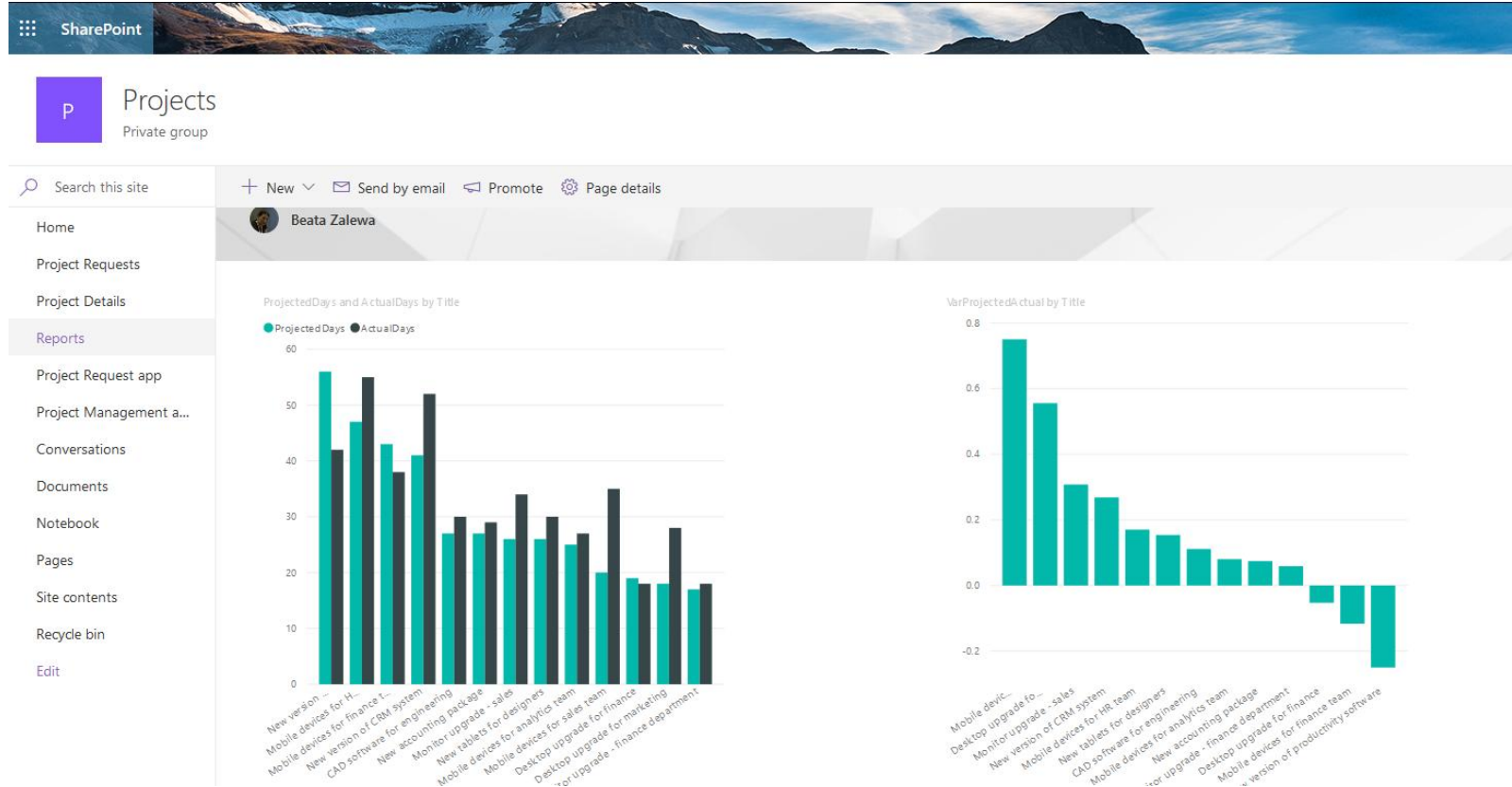
Embed link for SharePoint

Use the link below to securely embed this report in a SharePoint page. [Learn more](#)

<https://app.powerbi.com/reportEmbed?reportId=e69ffc96-740e-4006-8db4-dbc1c6e1d480&config=eyJjbH>

Close

Embed a report feature in Power BI



SharePoint page with reports

DEMO 5

SharePoint Online page with reports

<https://zalewa.sharepoint.com/sites/Projects/SitePages/Reports.aspx>

Q & A

Email: info@zalnet.pl

Resources:

<https://docs.microsoft.com>

Internet

Thank you for your precious time 😊